

Problem 01 — Forward difference at $x = 2.0$

Given table

x	2.0	2.2	2.4	2.6	2.8
f(x)	4.000	4.840	5.952	7.336	9.000

Step 1 — Identify h .

Spacing $h = 2.2 - 2.0 = 0.2$.

Step 2 — Write the forward difference formula (first order).

$$f'(x_0) \approx \frac{f(x_0 + h) - f(x_0)}{h}$$

Here $x_0 = 2.0$.

Step 3 — Substitute the values.

$$f'(2.0) \approx \frac{f(2.2) - f(2.0)}{0.2} = \frac{4.840 - 4.000}{0.2}$$

Step 4 — Compute numerator carefully (digit by digit).

$$4.840 - 4.000 = 0.840.$$

Now divide by 0.2:

$$\frac{0.840}{0.2} = 0.840 \div 0.2 = 4.200$$

Answer (boxed): $f'(2.0) \approx 4.20$

Problem 02 — Central difference at $x = 1.2$

Given table

x	0.8	1.0	1.2	1.4	1.6
f(x)	0.512	1.000	1.728	2.744	4.096

Step 1 — Identify h .

Spacing $h = 1.0 - 0.8 = 0.2$ (equally spaced).

Step 2 — Write the central difference formula.

For $x_i = 1.2$,

$$f'(x_i) \approx \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}.$$

Here $x_{i-1} = 1.0$, $x_{i+1} = 1.4$.

Step 3 — Substitute the values.

$$f'(1.2) \approx \frac{f(1.4) - f(1.0)}{2(0.2)} = \frac{2.744 - 1.000}{0.4}.$$

Step 4 — Compute numerator carefully.

$$2.744 - 1.000 = 1.744.$$

Divide by 0.4:

$$\frac{1.744}{0.4} = 1.744 \div 0.4 = 4.360.$$

Answer (boxed): $f'(1.2) \approx 4.360$